

Anemia in People with Cancer

Anemia is a condition that develops if your body is not making enough red blood cells (RBCs). Red blood cells have a protein called **hemoglobin** (Hgb), which carries oxygen throughout your body. Cells in your body need oxygen to function and survive.

[Anemia can be a sign of cancer](#)¹. If you have been diagnosed with cancer, anemia may be a side effect of cancer treatment, or it could be caused by the cancer itself.

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What causes anemia?

Low red blood cell counts (anemia) can be caused by cancer, cancer treatments, or something other than cancer. Some reasons people get anemia include:

- **Cancers that affect the bone marrow** (leukemias, lymphomas, multiple myeloma)
- **Blood loss** from injury or other problems
- **Cancer treatments** such as [radiation](#)² or [chemotherapy](#)³
- **Disorders that affect the bone marrow** such as [myelodysplastic syndromes](#)⁴ (MDS)
- **Bleeding tumors** (most common in the liver, stomach, and peritoneum)
- **Nutrition problems** such as low levels of iron, vitamin B12, or folate

- **Medicines** such as nonsteroidal anti-inflammatory drugs (NSAIDs), aspirin, steroids, antibiotics

Anemia symptoms that could occur during cancer

You might not feel any effects of anemia until your hemoglobin level is very low. Some of the most common signs and symptoms are:

- Extreme tiredness ([fatigue](#)⁵)
- Fast heartbeat
- Chest pain
- Shortness of breath
- Dizziness or lightheadedness
- [Swelling](#)⁶ in the hands or feet
- Pale skin, nail beds, mouth, or gums

Anemia can range from mild to severe. Certain health problems such as heart or lung problems can make anemia symptoms worse. Ask your cancer care team if your cancer or cancer treatment causes anemia.

Call 911 or go to the emergency room if you have

- Bleeding that won't stop
- New or worse confusion
- Shortness of breath even when resting
- Chest pain or irregular heartbeat

How is anemia found?

If you have signs or symptoms of anemia, your cancer care team will ask you to have a test called a complete blood count (CBC) to check your hemoglobin.

If you do have anemia, you might need other tests to try and find the cause. Some tests that might be done include:

- [Blood tests](#)⁷ to check your organs and levels of vitamins and minerals

- [Tests on body fluids](#)⁸ (such as stool, urine, sputum, or vomit) to check for blood
- A [bone marrow biopsy](#)⁹ to make sure your bone marrow is making healthy red blood cells
- [Imaging tests](#)¹⁰ (such as MRI or CAT scan) to look for bleeding in the body

How is anemia treated?

Treatments for low red blood cells depend on the cause. The most common treatments are:

- A [red blood cell transfusion](#)¹¹ if your hemoglobin is very low or if you have signs of bleeding
- Iron supplement therapy (given as an IV or pill)
- Medicines that tell the body to make new red blood cells (called erythropoiesis-stimulating agents or ESAs)
- Vitamin B12 or folic acid supplements (given as an IV, shot, or pill)

If chemotherapy, medicine, or other treatment is causing anemia, the doctor might lower the dose, switch to a different treatment, or stop the treatment altogether to give your body a chance to recover.

Tips for managing anemia during cancer

- If you have anemia caused by low iron, eat foods high in iron such as:
 - Red meat, fatty fish, chicken, and turkey
 - Dark leafy greens such as spinach, kale, collard greens, and chard
 - Beans, lentils, and tofu
 - Dried fruits such as raisins, apricots, and peaches
 - Fortified cereals
 - Enriched pasta and rice
- Drink plenty of water unless your doctor tells you to limit your fluids. It's OK to include other fluids such as broth and tea.
- When your red blood cells are low, you might feel tired, short of breath, or dizzy doing your normal activities. Be careful, ask for help, and rest when you need to.

Talk to your doctor or cancer care team if you

- Have dark brown or bright red vomit
- Have red or black stools
- Are dizzy, lightheaded, or have fallen
- Can't get out of bed for more than 24 hours

Hyperlinks

1. www.cancer.org/cancer/diagnosis-staging/signs-and-symptoms-of-cancer/anemia.html
2. www.cancer.org/cancer/managing-cancer/treatment-types/radiation.html
3. www.cancer.org/cancer/managing-cancer/treatment-types/chemotherapy.html
4. www.cancer.org/cancer/types/myelodysplastic-syndrome.html
5. www.cancer.org/cancer/managing-cancer/side-effects/fatigue-weakness-sleep.html
6. www.cancer.org/cancer/managing-cancer/side-effects/swelling.html
7. www.cancer.org/cancer/diagnosis-staging/tests/understanding-your-lab-test-results.html
8. www.cancer.org/cancer/diagnosis-staging/tests/biopsy-and-cytology-tests.html
9. www.cancer.org/cancer/diagnosis-staging/tests/biopsy-and-cytology-tests.html
10. www.cancer.org/cancer/diagnosis-staging/tests/imaging-tests/imaging-radiology-tests-for-cancer.html
11. www.cancer.org/cancer/managing-cancer/treatment-types/blood-transfusion-and-donation.html

References

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